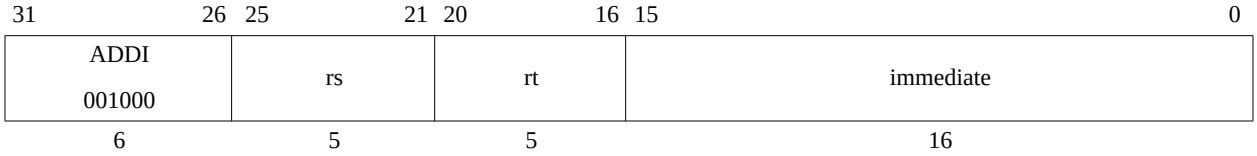
Add Immediate Word — ADDI



Format: `ADDI rt, rs, immediate` **MIPS32 (MIPS I)**

Purpose:

To add a constant to a 32-bit integer. If overflow occurs, then trap**.**

Description: `rt ← rs + immediate`

The 16-bit signed *immediate* is added to the 32-bit value in GPR *rs* to produce a 32-bit result.

- If the addition results in 32-bit 2's complement arithmetic overflow, the destination register is not modified and an Integer Overflow exception occurs.
- If the addition does not overflow, the 32-bit result is placed into GPR *rt*.

Restrictions:

None

Operation:

```
temp ← (GPR[rs]31||GPR[rs]31..0) + sign_extend(immediate)
if temp32 ≠ temp31 then
   SignalException(IntegerOverflow)
else
   GPR[rt] ← temp
endif
```

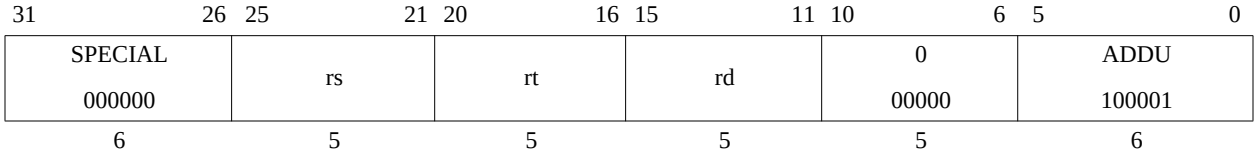
Exceptions:

Integer Overflow

Programming Notes:

ADDIU performs the same arithmetic operation but does not trap on overflow.

Add Unsigned Word — ADDU



Format: `ADDU rd, rs, rt` **MIPS32 (MIPS I)**

Purpose:

To add 32-bit integers

Description: `rd ← rs + rt`

The 32-bit word value in GPR *rt* is added to the 32-bit value in GPR *rs* and the 32-bit arithmetic result is placed into GPR *rd*.

No Integer Overflow exception occurs under any circumstances.

Restrictions:

None

Operation:

```
temp ← GPR[rs] + GPR[rt]
GPR[rd] ← temp
```

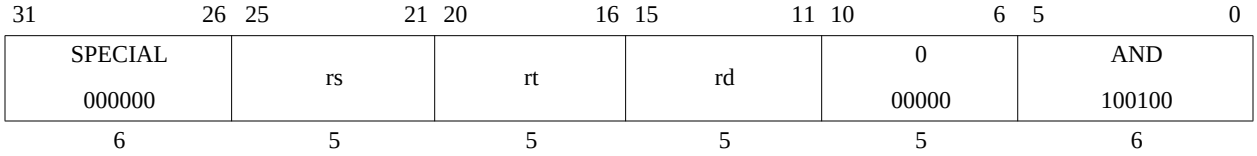
Exceptions:

None

Programming Notes:

The term “unsigned” in the instruction name is a misnomer; this operation is 32-bit modulo arithmetic that does not trap on overflow. This instruction is appropriate for unsigned arithmetic, such as address arithmetic, or integer arithmetic environments that ignore overflow, such as C language arithmetic.

And AND



Format: `AND rd, rs, rt` **MIPS32 (MIPS I)**

Purpose:

To do a bitwise logical AND

Description: `rd ← rs AND rt`

The contents of GPR *rs* are combined with the contents of GPR *rt* in a bitwise logical AND operation. The result is placed into GPR *rd*.

Restrictions:

None

Operation:

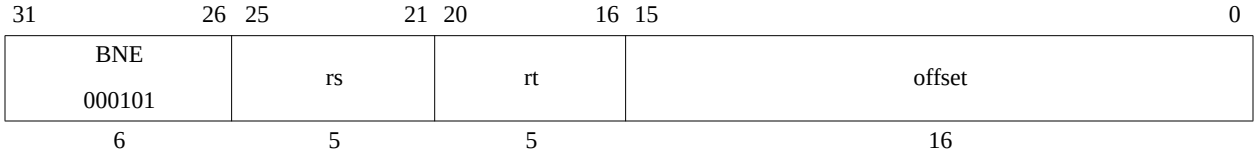
```
GPR[rd] ← GPR[rs] and GPR[rt]
```

Exceptions:

None

Branch on Not Equal

BNE



Format: `BNE rs, rt, offset` **MIPS32 (MIPS I)**

Purpose:

To compare GPRs then do a PC-relative conditional branch

Description: `if rs ≠ rt then branch`

An 18-bit signed offset (the 16-bit *offset* field shifted left 2 bits) is added to the address of the instruction following the branch (not the branch itself), in the branch delay slot, to form a PC-relative effective target address.

If the contents of GPR *rs* and GPR *rt* are not equal, branch to the effective target address after the instruction in the delay slot is executed.

Restrictions:

Processor operation is **UNPREDICTABLE** if a branch, jump, ERET, DERET, or WAIT instruction is placed in the delay slot of a branch or jump.

Operation:

```
I:     target_offset ← sign_extend(offset || 0^2)
       condition ← (GPR[rs] ≠ GPR[rt])
I+1:   if condition then
          PC ← PC + target_offset
       endif
```

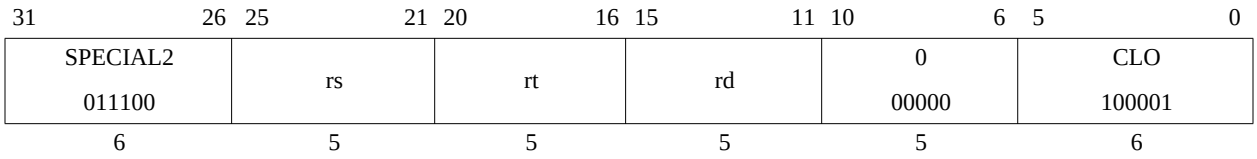
Exceptions:

None

Programming Notes:

With the 18-bit signed instruction offset, the conditional branch range is ± 128 KBytes. Use jump (J) or jump register (JR) instructions to branch to addresses outside this range.

Count Leading Ones in Word — CLO



Format: `CLO rd, rs` MIPS32

Purpose:

To Count the number of leading ones in a word

Description: `rd ← count_leading_ones rs`

Bits 31..0 of GPR *rs* are scanned from most significant to least significant bit. The number of leading ones is counted and the result is written to GPR *rd*. If all of bits 31..0 were set in GPR *rs*, the result written to GPR *rd* is 32.

Restrictions:

To be compliant with the MIPS32 and MIPS64 Architecture, software must place the same GPR number in both the *rt* and *rd* fields of the instruction. The operation of the instruction is **UNPREDICTABLE** if the *rt* and *rd* fields of the instruction contain different values.

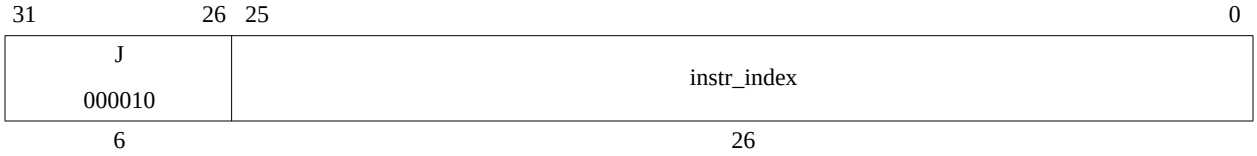
Operation:

```
temp ← 32
for i in 31 .. 0
   if GPR[rs]i = 0 then
      temp ← 31 - i
      break
   endif
endfor
GPR[rd] ← temp
```

Exceptions:

None

Jump J



Format: `J target` **MIPS32 (MIPS I)**

Purpose:

To branch within the current 256 MB-aligned region

Description:

This is a PC-region branch (not PC-relative); the effective target address is in the “current” 256 MB-aligned region. The low 28 bits of the target address is the *instr_index* field shifted left 2 bits. The remaining upper bits are the corresponding bits of the address of the instruction in the delay slot (not the branch itself).

Jump to the effective target address. Execute the instruction that follows the jump, in the branch delay slot, before executing the jump itself.

Restrictions:

Processor operation is **UNPREDICTABLE** if a branch, jump, ERET, DERET, or WAIT instruction is placed in the delay slot of a branch or jump.

Operation:

```
I:
I+1:PC ← PC_GPRLEN..28 || instr_index || 0^2
```

Exceptions:

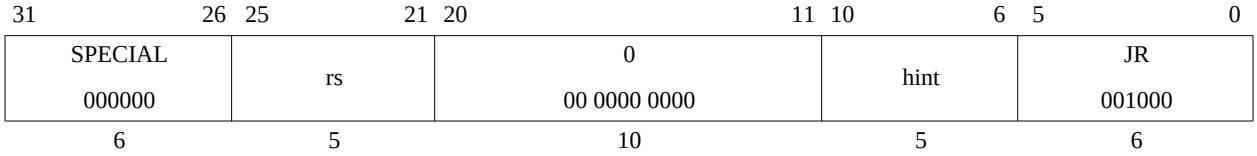
None

Programming Notes:

Forming the branch target address by catenating PC and index bits rather than adding a signed offset to the PC is an advantage if all program code addresses fit into a 256 MB region aligned on a 256 MB boundary. It allows a branch from anywhere in the region to anywhere in the region, an action not allowed by a signed relative offset.

This definition creates the following boundary case: When the jump instruction is in the last word of a 256 MB region, it can branch only to the following 256 MB region containing the branch delay slot.

Jump Register JR



Format: `JR rs` **MIPS32 (MIPS I)**

Purpose:

To execute a branch to an instruction address in a register

Description: `PC ← rs`

Jump to the effective target address in GPR *rs*. Execute the instruction following the jump, in the branch delay slot, before jumping.

For processors that implement the MIPS16 ASE, set the *ISA Mode* bit to the value in GPR *rs* bit 0. Bit 0 of the target address is always zero so that no Address Exceptions occur when bit 0 of the source register is one

Restrictions:

The effective target address in GPR *rs* must be naturally-aligned. For processors that do not implement the MIPS16 ASE, if either of the two least-significant bits are not zero, an Address Error exception occurs when the branch target is subsequently fetched as an instruction. For processors that do implement the MIPS16 ASE, if bit 0 is zero and bit 1 is one, an Address Error exception occurs when the jump target is subsequently fetched as an instruction.

At this time the only defined hint field value is 0, which sets default handling of JR. Future versions of the architecture may define additional hint values.

Processor operation is **UNPREDICTABLE** if a branch, jump, ERET, DERET, or WAIT instruction is placed in the delay slot of a branch or jump.

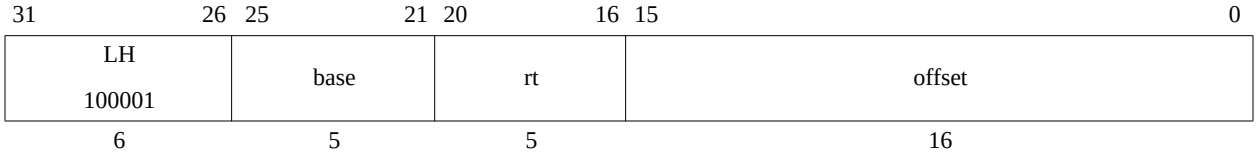
Operation:

```
I:  temp ← GPR[rs]
I+1:if Config1CA = 0 then
       PC ← temp
    else
       PC ← tempGPRLEN-1..1 || 0
       ISAMode ← temp0
    endif
```

Exceptions:

None

Load Halfword LH



Format: `LH rt, offset(base)` **MIPS32 (MIPS I)**

Purpose:

To load a halfword from memory as a signed value

Description: `rt ← memory[base+offset]`

The contents of the 16-bit halfword at the memory location specified by the aligned effective address are fetched, sign-extended, and placed in GPR *rt*. The 16-bit signed *offset* is added to the contents of GPR *base* to form the effective address.

Restrictions:

The effective address must be naturally-aligned. If the least-significant bit of the address is non-zero, an Address Error exception occurs.

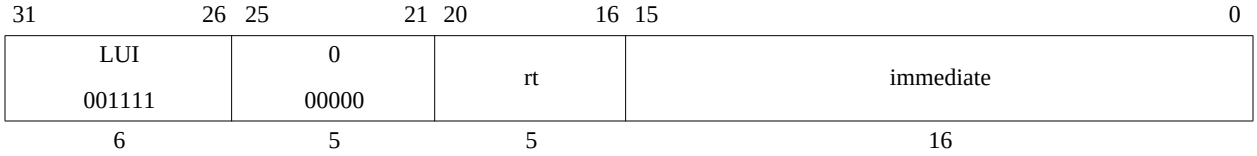
Operation:

```
vAddr  ← sign_extend(offset) + GPR[base]
if vAddr0 ≠ 0 then
   SignalException(AddressError)
endif
(pAddr, CCA) ← AddressTranslation (vAddr, DATA, LOAD)
pAddr  ← pAddrPSIZE-1..2 || (pAddr1..0 xor (ReverseEndian || 0))
memword ← LoadMemory (CCA, HALFWORD, pAddr, vAddr, DATA)
byte   ← vAddr1..0 xor (BigEndianCPU || 0)
GPR[rt] ← sign_extend(memword15+8*byte..8*byte)
```

Exceptions:

TLB Refill, TLB Invalid, Bus Error, Address Error

Load Upper Immediate — LUI



Format: `LUI rt, immediate` **MIPS32 (MIPS I)**

Purpose:

To load a constant into the upper half of a word

Description: `rt ← immediate || 0^16`

The 16-bit *immediate* is shifted left 16 bits and concatenated with 16 bits of low-order zeros. The 32-bit result is placed into GPR *rt*.

Restrictions:

None

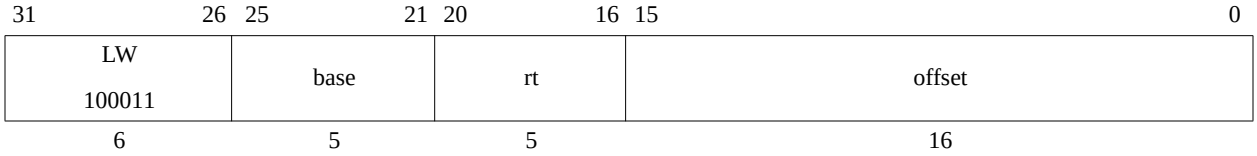
Operation:

```
GPR[rt] ← immediate || 0^16
```

Exceptions:

None

Load Word LW



Format: `LW rt, offset(base)` **MIPS32 (MIPS I)**

Purpose:

To load a word from memory as a signed value

Description: `rt ← memory[base+offset]`

The contents of the 32-bit word at the memory location specified by the aligned effective address are fetched, sign-extended to the GPR register length if necessary, and placed in GPR *rt*. The 16-bit signed *offset* is added to the contents of GPR *base* to form the effective address.

Restrictions:

The effective address must be naturally-aligned. If either of the 2 least-significant bits of the address is non-zero, an Address Error exception occurs.

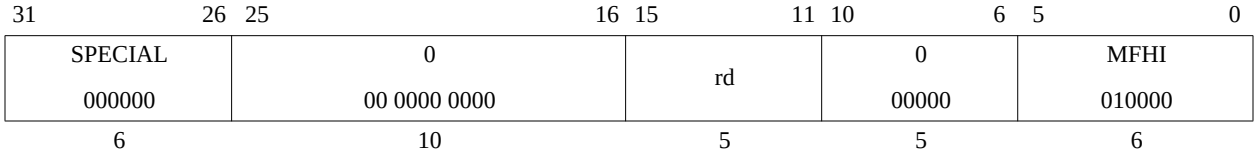
Operation:

```
vAddr ← sign_extend(offset) + GPR[base]
if vAddr1..0 ≠ 02 then
   SignalException(AddressError)
endif
(pAddr, CCA)← AddressTranslation (vAddr, DATA, LOAD)
memword← LoadMemory (CCA, WORD, pAddr, vAddr, DATA)
GPR[rt]← memword
```

Exceptions:

TLB Refill, TLB Invalid, Bus Error, Address Error

Move From HI Register MFHI



Format: `MFHI rd` **MIPS32 (MIPS I)**

Purpose:

To copy the special purpose *HI* register to a GPR

Description: `rd ← HI`

The contents of special register *HI* are loaded into GPR *rd*.

Restrictions:

None

Operation:

```
GPR[rd] ← HI
```

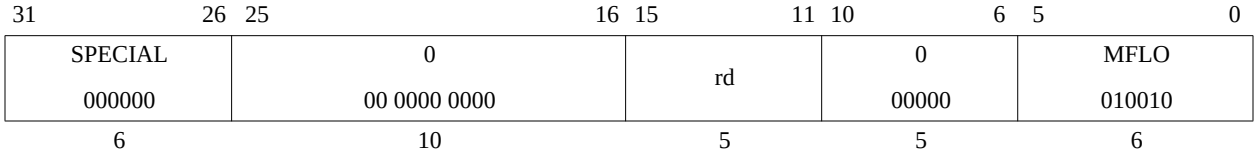
Exceptions:

None

Historical Information:

In the MIPS I, II, and III architectures, the two instructions which follow the MFHI must not moodify the HI register. If this restriction is violated, the result of the MFHI is **UNPREDICTABLE**. This restriction was removed in MIPS IV and MIPS32, and all subsequent levels of the architecture.

Move From LO Register MFLO



Format: `MFLO rd` **MIPS32 (MIPS I)**

Purpose:

To copy the special purpose *LO* register to a GPR

Description: `rd ← LO`

The contents of special register *LO* are loaded into GPR *rd*.

Restrictions: None

Operation:

```
GPR[rd] ← LO
```

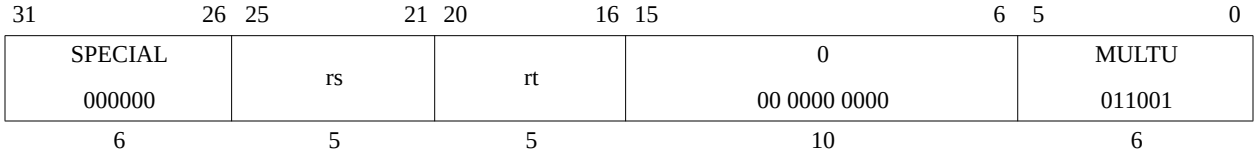
Exceptions:

None

Historical Information:

In the MIPS I, II, and III architectures, the two instructions which follow the MFHI must not moodify the HI register. If this restriction is violated, the result of the MFHI is **UNPREDICTABLE**. This restriction was removed in MIPS IV and MIPS32, and all subsequent levels of the architecture.

Multiply Unsigned Word — MULTU



Format: `MULTU rs, rt` **MIPS32 (MIPS I)**

Purpose:

To multiply 32-bit unsigned integers

Description: `(LO, HI) ← rs × rt`

The 32-bit word value in GPR *rt* is multiplied by the 32-bit value in GPR *rs*, treating both operands as unsigned values, to produce a 64-bit result. The low-order 32-bit word of the result is placed into special register *LO*, and the high-order 32-bit word is placed into special register *HI*.

No arithmetic exception occurs under any circumstances.

Restrictions:

None

Operation:

```
prod← (0 || GPR[rs]31..0) × (0 || GPR[rt]31..0)
LO ← prod31..0
HI ← prod63..32
```

Exceptions:

None

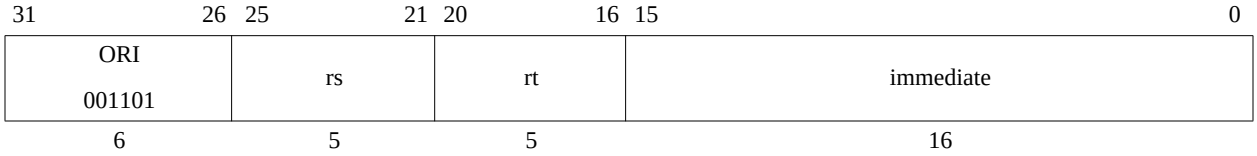
Programming Notes:

In some processors the integer multiply operation may proceed asynchronously and allow other CPU instructions to execute before it is complete. An attempt to read *LO* or *HI* before the results are written interlocks until the results are ready. Asynchronous execution does not affect the program result, but offers an opportunity for performance improvement by scheduling the multiply so that other instructions can execute in parallel.

Programs that require overflow detection must check for it explicitly.

Where the size of the operands are known, software should place the shorter operand in GPR *rt*. This may reduce the latency of the instruction on those processors which implement data-dependent instruction latencies.

Or Immediate ORI



Format: `ORI rt, rs, immediate` **MIPS32 (MIPS I)**

Purpose:

To do a bitwise logical OR with a constant

Description: `rt ← rs or immediate`

The 16-bit *immediate* is zero-extended to the left and combined with the contents of GPR *rs* in a bitwise logical OR operation. The result is placed into GPR *rt*.

Restrictions:

None

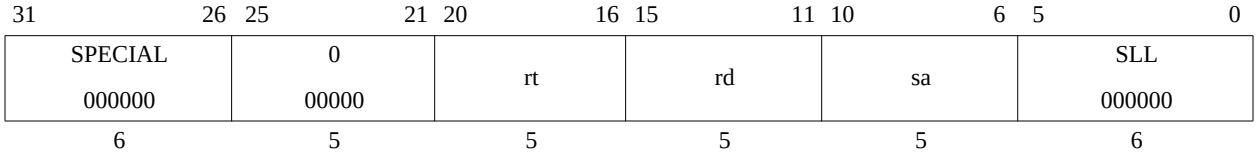
Operation:

```
GPR[rt] ← GPR[rs] or zero_extend(immediate)
```

Exceptions:

None

Shift Word Left Logical — SLL



Format: `SLL rd, rt, sa` **MIPS32 (MIPS I)**

Purpose:

To left-shift a word by a fixed number of bits

Description: `rd ← rt << sa`

The contents of the low-order 32-bit word of GPR *rt* are shifted left, inserting zeros into the emptied bits; the word result is placed in GPR *rd*. The bit-shift amount is specified by *sa*.

Restrictions:

None

Operation:

```
s      ← sa
temp   ← GPR[rt](31-s)..0 || 0^s
GPR[rd]← temp
```

Exceptions:

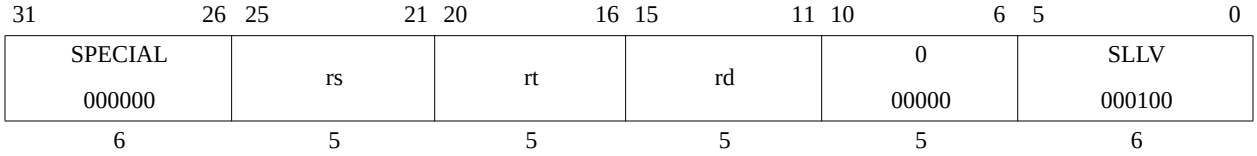
None

Programming Notes:

SLL r0, r0, 0, expressed as NOP, is the assembly idiom used to denote no operation.

SLL r0, r0, 1, expressed as SSNOP, is the assembly idiom used to denote no operation that causes an issue break on superscalar processors.

Shift Word Left Logical Variable — SLLV



Format: `SLLV rd, rt, rs` **MIPS32 (MIPS I)**

Purpose: To left-shift a word by a variable number of bits

Description: `rd ← rt << rs`

The contents of the low-order 32-bit word of GPR *rt* are shifted left, inserting zeros into the emptied bits; the result word is placed in GPR *rd*. The bit-shift amount is specified by the low-order 5 bits of GPR *rs*.

Restrictions: None

Operation:

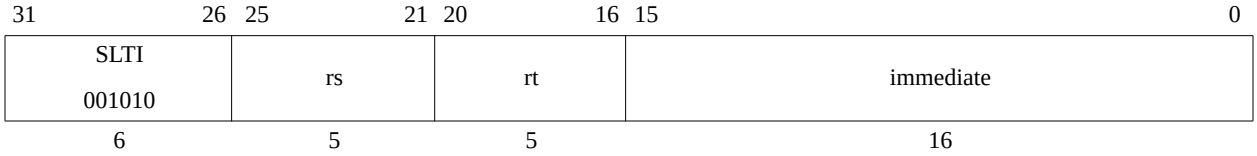
```
s      ← GPR[rs]4..0
temp   ← GPR[rt](31-s)..0 || 0s
GPR[rd]← temp
```

Exceptions: None

Programming Notes:

None

Set on Less Than Immediate SLTI



Format: `SLTI rt, rs, immediate` **MIPS32 (MIPS I)**

Purpose:

To record the result of a less-than comparison with a constant

Description: `rt ← (rs < immediate)`

Compare the contents of GPR *rs* and the 16-bit signed *immediate* as signed integers and record the Boolean result of the comparison in GPR *rt*. If GPR *rs* is less than *immediate*, the result is 1 (true); otherwise, it is 0 (false).

The arithmetic comparison does not cause an Integer Overflow exception.

Restrictions:

None

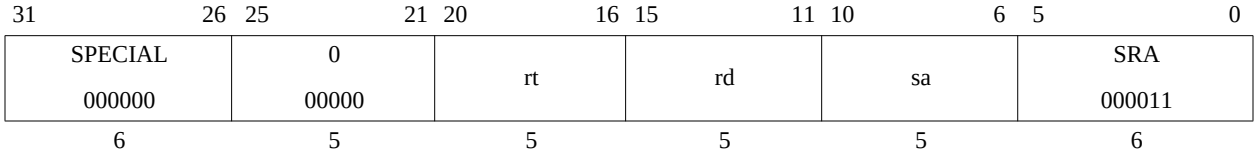
Operation:

```
if GPR[rs] < sign_extend(immediate) then
   GPR[rd] ← 0^(GPRLEN-1)|| 1
else
   GPR[rd] ← 0^GPRLEN
endif
```

Exceptions:

None

Shift Word Right Arithmetic SRA



Format: `SRA rd, rt, sa` **MIPS32 (MIPS I)**

Purpose:

To execute an arithmetic right-shift of a word by a fixed number of bits

Description: `rd ← rt >> sa (arithmetic)`

The contents of the low-order 32-bit word of GPR *rt* are shifted right, duplicating the sign-bit (bit 31) in the emptied bits; the word result is placed in GPR *rd*. The bit-shift amount is specified by *sa*.

Restrictions:

None

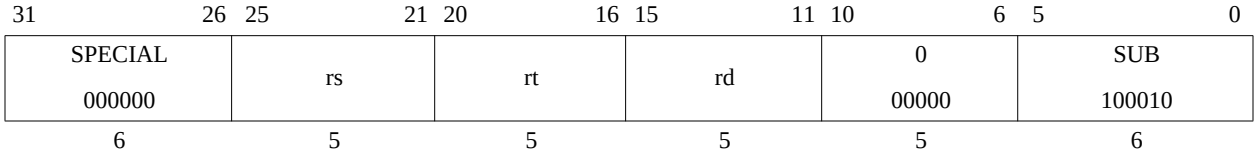
Operation:

```
s      ← sa
temp   ← (GPR[rt]31)^s || GPR[rt]31..s
GPR[rd]← temp
```

Exceptions: None

Subtract Word

SUB



Format: `SUB rd, rs, rt` **MIPS32 (MIPS I)**

Purpose:

To subtract 32-bit integers. If overflow occurs, then trap

Description: `rd ← rs - rt`

The 32-bit word value in GPR *rt* is subtracted from the 32-bit value in GPR *rs* to produce a 32-bit result. If the subtraction results in 32-bit 2's complement arithmetic overflow, then the destination register is not modified and an Integer Overflow exception occurs. If it does not overflow, the 32-bit result is placed into GPR *rd*.

Restrictions:

None

Operation:

```
temp ← (GPR[rs]31||GPR[rs]31..0) – (GPR[rt]31||GPR[rt]31..0)
if temp32 ≠ temp31 then
   SignalException(IntegerOverflow)
else
   GPR[rd] ← temp31..0
endif
```

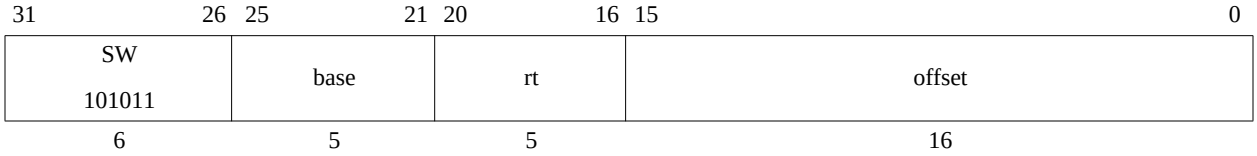
Exceptions:

Integer Overflow

Programming Notes:

SUBU performs the same arithmetic operation but does not trap on overflow.

Store Word SW



Format: `SW rt, offset(base)` **MIPS32 (MIPS I)**

Purpose:

To store a word to memory

Description: `memory[base+offset] ← rt`

The least-significant 32-bit word of register *rt* is stored in memory at the location specified by the aligned effective address. The 16-bit signed *offset* is added to the contents of GPR *base* to form the effective address.

Restrictions:

The effective address must be naturally-aligned. If either of the 2 least-significant bits of the address is non-zero, an Address Error exception occurs.

Operation:

```
vAddr ← sign_extend(offset) + GPR[base]
if vAddr1..0 ≠ 02 then
   SignalException(AddressError)
endif
(pAddr, CCA)← AddressTranslation (vAddr, DATA, STORE)
dataword← GPR[rt]
StoreMemory (CCA, WORD, dataword, pAddr, vAddr, DATA)
```

Exceptions:

TLB Refill, TLB Invalid, TLB Modified, Address Error